MATH1141: Higher Mathematics 1A (Calculus)

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0.1 Content Overview

- Functions: continuity, differentiability, invertibility (inverse functions)
- Curve sketching
- Integration: area, Riemann sum (approximation), Fundamental theorems of calculus
- Logarithm: exponential, hyperbolic

Chapter 1

Sets, inequalities and functions

1.1 Sets of numbers

A set is a collection of distinct objects. The objects in a set are elements or members of the set. Commonly used sets:

- The set of natural numbers: $\mathbb{N} = \{0, 1, 2, 3, 4, ...\}$
- The set of integers: $\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$
- The set of rational numbers: $\mathbb{Q} = \{\frac{p}{q} : p, q$ are integers and $q \neq 0\}$
- The set of $\mathbb{R}$ real numbers may be represented as the collection of points lying on the number line

If A is a set of numbers and the number x is a member of the set A, then we write:

$$x \in A$$

If x is not a member of A, then we write

$$x \notin A$$

1.1.1 Intervals

- Parenthesis: excludes endpoints
- Bracket: includes endpoints
- Combination: neither open nor closed interval

1.1.2 Solving Inequalities

For $x, y, z \in \mathbb{R}$:

- if $x > y$, then $x + z > y + z$
- if $x > y$ and $z > 0$, then $xz > yz$
- if $x > y$ and $z < 0$, then $xz < yz$

Example

Solve the quadratic inequality $x^2 + 4x > 21$

$$\begin{aligned} x^2 + 4x > 21 &\iff x^2 + 4x - 21 > 0 \\ &\iff (x+7)(x-3) > 0 \\ \text{Solution set: } x < -7 \text{ or } x > 3 & \\ x \in (-\infty, -7) \cup (3, \infty) & \end{aligned}$$

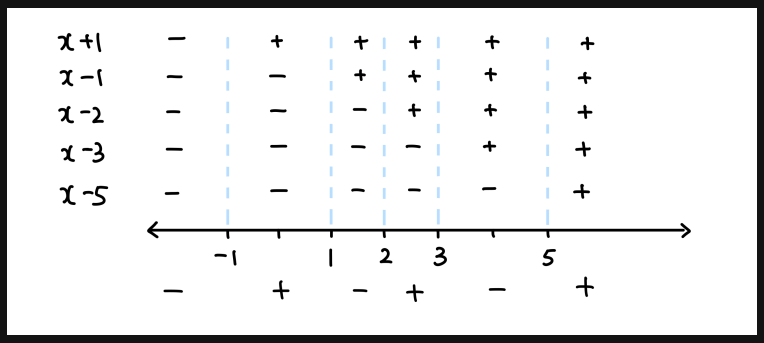
Example

Solve the rational inequality $\frac{1}{x+1} < \frac{1}{(x-2)(x-2)}$

Multiply both sides by $(x+1)^2(x-2)^2(x-3)^2 \quad (>= 0)$,

$$\begin{aligned} \frac{1}{x+1} < \frac{1}{(x-2)(x-3)} &\iff (x+1)(x-2)^2(x-3)^2 < (x+1)^2(x-2)(x-3) \\ &\iff (x+1)(x-2)^2(x-3)^2 - (x+1)^2(x-2)(x-3) < 0 \\ &\iff (x+1)(x-2)(x-3)(x-4)(x-5) < 0 \end{aligned}$$

Solution set: $(-\infty, -1) \cup (1, 2) \cup (3, 5)$



1.1.3 Absolute Values

The absolute value (or magnitude) of a real number x is

$$|x| = \begin{cases} x & \text{if } x \le 0, \\ -x & \text{if } x < 0 \end{cases}$$

Example

Solve the inequality $|3x+1| \le 4$

$$\begin{aligned} |3x+1| \le 4 &\iff |x + \frac{1}{3}| \le \frac{4}{3} \\ &\iff x + \frac{1}{3} \le \frac{-4}{3} \quad \text{or} \quad x + \frac{1}{3} \le \frac{4}{3} \\ &\iff x \le \frac{-5}{3} \quad \text{or} \quad x \le 1 \end{aligned}$$

Solution set: $(-\infty, \frac{-5}{3}] \cup [1, \infty)$

Example

Solve the inequality $\frac{|x+5|}{|x-11|} < 1$

$$\begin{aligned}\frac{|x+5|}{|x-11|} < 1 &\iff |x+5| < |x-11| \quad \text{(multiply both sides by } |x\text{-}11|\text{)}\\ &\iff |x+5|^2 < |x-11|^2\\ &\iff x^2+10x+25 < x^2-22x+121\\ &\iff 32x < 96\\ &\iff x < 3\end{aligned}$$

Solution set: $(-\infty, 3)$

Note: 11 is not in the solution set

1.1.4 Functions

Domain, codomain and range

Let A and B be subsets of $\mathbb{R}$: A function with domain A and codomain B is a rule which assigns to every $x \in A$ exactly one number $f(x) \in B$

$$\begin{aligned}&f : A \to B\\ &f : A \ni x \mapsto f(x) \in B\end{aligned}$$

- $A = \text{Dom}(f)$: set of all inputs
- $B = \text{Codom}(f)$: set that contains all the outputs
- Range of f: set of actual outputs, defined by:

$$\text{Range}(f) = \{f(x) \in B : x \in A\}$$

- Hence, $\text{Range}(f) \subseteq \text{Codom}(f)$
- The codomain does not have to be unique
- The codomain is useful when the range is unknown

Example

Given: $f : [0,\infty) \to \mathbb{R}, f(x) = 2 + \sqrt{x}, \forall x \in [0,\infty)$
Then, $\text{Dom}(f) = [0,\infty)$, $\text{Codom}(f) = \mathbb{R}$, and $\text{Range}(f) = [2,\infty)$

Natural domain (or maximal domain) of f is the largest possible domain for which the rule makes sense (for real numbers)

Forming new functions

Let A, B, C and D be subsets of $\mathbb{R}$. Given two functions $f : A \to B$ and $g : A \to B$, then the functions

$$f+g, \quad f-g, \quad fg, \quad \frac{f}{g}$$

are defined by the rules

$$
\begin{aligned}
(f+g)(x) &= f(x)+g(x), \quad \forall x \in A \\
(f-g)(x) &= f(x)-g(x), \quad \forall x \in A \\
(fg)(x) &= f(x)g(x), \quad \forall x \in A \\
\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}, \quad \forall x \in \mathrm{A} \text{ provided that } g(x) = 0.
\end{aligned}
$$

To form these new functions, the domains of both f and g must be the same

Suppose that $f : C \to D$ and $g : A \to B$ are functions such that

$$\mathrm{Range}(g) \subseteq \mathrm{Dom}(f) = C$$

Then the composition function $(f \circ g)(x) = f(g(x)), \quad \forall x \in A$

1.1.5 The Elementary Functions

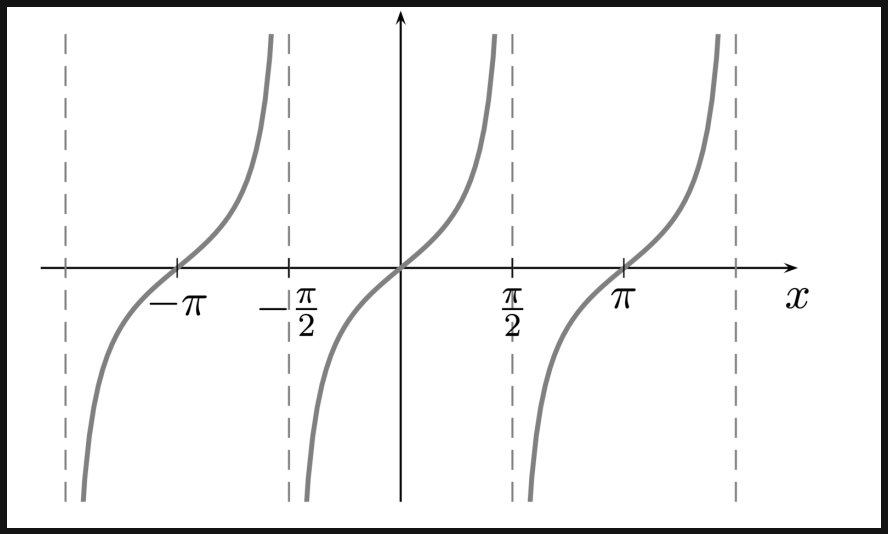
Elementary functions are those that can be constructed by combining a finite number of polynomials, exponentials, logarithms, roots, trigonometric and inverse trigonometric functions via function composition ($\circ$), $+$, $-$, $\times$, and $\div$.

1.1.6 Implicitly Defined Functions

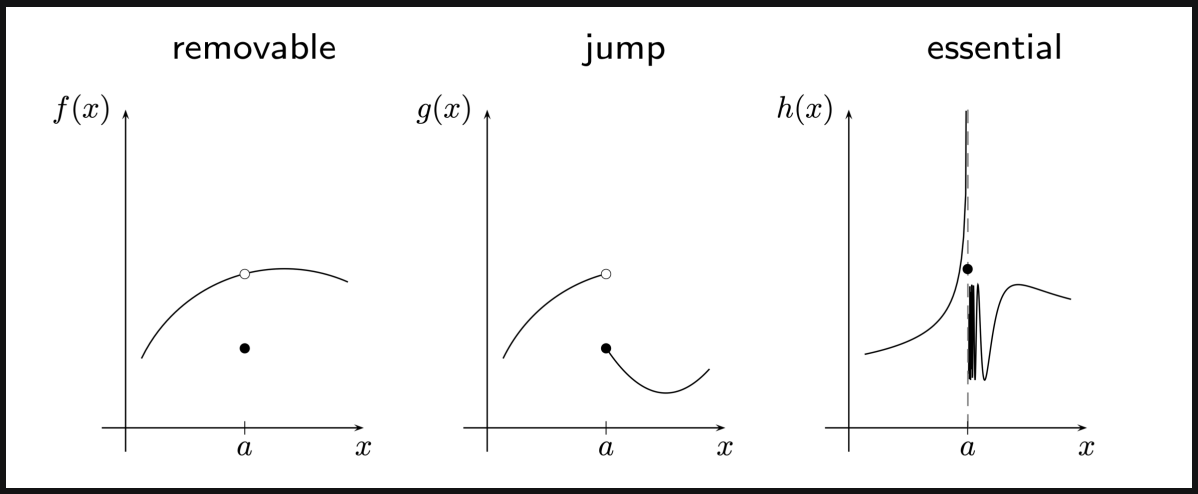
Many curves on the plane can be described by the points (x, y) that satisfy some equation involving x and y. However, not all these curves are necessarily functions. Some cannot be expressed with a single y term ($y = \ldots$). In this case, the curve may be decomposed into two or more functions that are implicitly defined by the curve.

1.1.7 Continuous Functions

In the below example, the $\tan x$ graph breaks at $x = -\frac{\pi}{2}, \frac{\pi}{2}, \ldots$.



However, this break in the domain does not make it a discontinuity in the graph. Discontinuities can be categorised as following:


removable
jump
essential
$f(x)$
$g(x)$
$h(x)$
a
x
a
x
a
x

Chapter 2

Limits

2.1 Limits of functions at infinity

- If $f(x)$ gets arbitrarily close to some $L \in \mathbb{R}$ as x tends to ∞, then we write:

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad f(x) \to L \text{ as } x \to L$$

- If $f(x)$ gets arbitrarily large as x tends to ∞, then we write:

$$f(x) \to \infty \text{ as } x \to \infty$$

  - In this case, we state that the limit of $f(x)$ as x tends to ∞ does not exist

$$\lim_{x\to\infty} \frac{1}{f(x)} = 0$$

- The limit notation can only be used when the limit has a constant definition (do not write $\lim_{x\to\infty} = \infty$)

2.1.1 Basic rules for limits

If $\lim_{x\to\infty} f(x) = L_1$ and $\lim_{x\to\infty} g(x) = L_2$ then

- $\lim_{x\to\infty} (f(x) + g(x)) = L_1 + L_2$
- $\lim_{x\to\infty} (f(x) - g(x)) = L_1 - L_2$
- $\lim_{x\to\infty} f(x)g(x) = L_1 L_2$
- $\lim_{x\to\infty} \dfrac{f(x)}{g(x)} = \dfrac{L_1}{L_2}$, provided that $L_2 \neq 0$

Example

Find the limit of $f(x) = \frac{1+\frac{2}{3x+4}}{5-6e^{-x}}$ as $x \to \infty$ (if it exists)

$$\begin{aligned}
\lim_{x\to\infty} f(x) &= \frac{\lim\limits_{x\to\infty}(1+\dfrac{2}{3x+4})}{\lim\limits_{x\to\infty}(5-6e^{-x})} \\
&= \frac{\lim\limits_{x\to\infty} 1 + \lim\limits_{x\to\infty} \dfrac{2}{3x+4}}{\lim\limits_{x\to\infty} 5 - \lim\limits_{x\to\infty} 6e^{-x}} \\
&= \frac{1+0}{5-0} \\
&= \frac{1}{5}
\end{aligned}$$

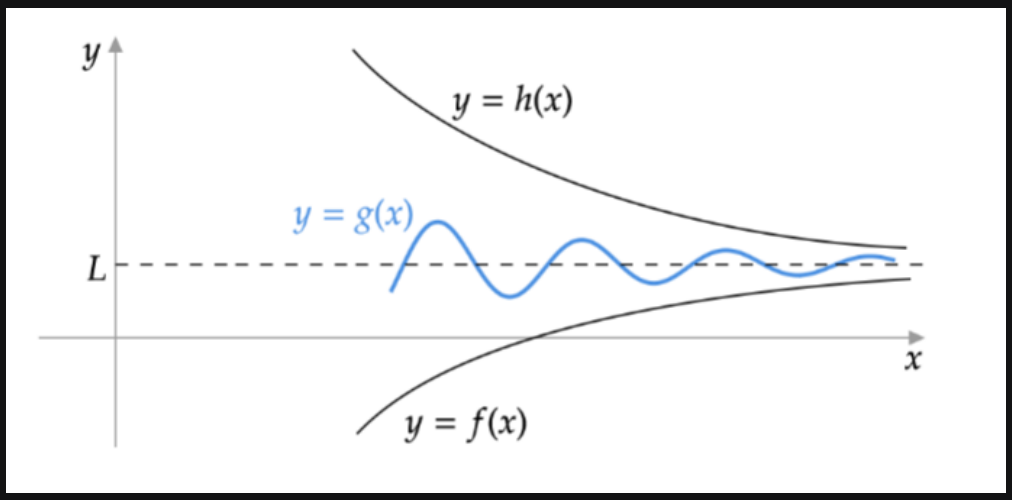
2.1.2 The pinching theorem

Suppose that f and h have the same limit as $x \to \infty$. If the graph of g always lies between the graphs of f and h, then g has the same limit.

Theorem Let f, g, and h be functions defined on (b,∞) for some $b \in \mathbb{R}.If$:

- $f(x) \le g(x) \le h(x), \quad \forall x \in (b,\infty)$, and
- $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} h(x) = L$, then

$$\lim_{x\to\infty} g(x) = L$$



Example

Find the limit of $g(x) = e^{-x^2} 2^{\sin 3x}$ as $x \to \infty$ (if it exists).

$$\begin{aligned}
&\sin x \in [-1, 1], \quad \forall x \in \mathbb{R} \\
&\Rightarrow \frac{1}{2} \leq 2^{\sin 3x} \leq 2, \quad \forall x \in \mathbb{R} \\
&\Rightarrow \frac{e^{-x^2}}{2} \leq e^{x^2} 2^{\sin 3x} \leq 2e^{-x^2} \\
\text{Since } &\lim_{x\to\infty} \frac{e^{-x^2}}{2} = \lim_{x\to\infty} 2e^{-x^2} = 0, \\
&\therefore \lim_{x\to\infty} e^{-x^2} 2^{\sin 3x} = 0
\end{aligned}$$

2.1.3 Limits of the form $f(x)/g(x)$

When calculating $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ when both $f(x)$ and $g(x)$ tend to ∞ as $x \to \infty$, the previous rules cannot be applied. The main goal in this situation is to **divide both f and g by the fastest growing term in g**.

Example

Find the limit $\lim\limits_{x\to\infty} \dfrac{5x^3 + 6x^2 - 4\sin x}{\cos 3x + 5x - 2x^3}$ (if it exists).

$$\begin{aligned}
\frac{5x^3 + 6x^2 - 4\sin x}{\cos 3x + 5x - 2x^3} &= \frac{5 + \frac{6}{x} - 4\frac{\sin x}{x^3}}{\frac{\cos 3x}{x^3} + \frac{5}{x^2} - 2} \\
&= \frac{5 + 0 + 0}{0 + 0 - 2} \\
&= -\frac{5}{2} \quad \text{as } x \to \infty
\end{aligned}$$

Example

Find the limit $\lim\limits_{x\to\infty} \dfrac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x}$ (if it exists).

$$\begin{aligned}
\frac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x} &= \frac{1 + \frac{3}{x}}{\frac{1}{x^2}\sqrt{2x^4 + 3} - \frac{4}{x}} \\
&= \frac{1 + \frac{3}{x}}{\sqrt{2 + \frac{3}{x^4}} - \frac{4}{x}} \\
&\to \frac{1 + 0}{\sqrt{2 + 0} - 0} \\
&= \frac{1}{\sqrt{2}} \quad \text{as } x \to \infty
\end{aligned}$$

2.1.4 Limits of the form $\sqrt{f(x)} - \sqrt{g(x)}$

In this case, the leading term may not help. Instead the lower order term will usually determine the limit. This can be achieved by **multiplying the numerator and denominator by the conjugate**.

Example

Find the limit of $\lim_{x\to\infty} \sqrt{2x^2+3x} - \sqrt{2x^2-x}$.

$$\begin{aligned}
\sqrt{2x^2+3x} - \sqrt{2x^2-x} &= \frac{\left(\sqrt{2x^2+3x} - \sqrt{2x^2-x}\right)\left(\sqrt{2x^2+3x} + \sqrt{2x^2-x}\right)}{\left(\sqrt{2x^2+3x} + \sqrt{2x^2-x}\right)} \\
&= \frac{2x^2+3x-\left(2x^2-x\right)}{\sqrt{2x^2+3x}+\sqrt{2x^2-x}} \\
&= \frac{4x}{\sqrt{2x^2+3x}+\sqrt{2x^2-x}} \\
&= \frac{x}{\sqrt{2+\frac{3}{x}}+\sqrt{2-\frac{1}{x}}} \\
&\to \frac{4}{\sqrt{2}+\sqrt{2}} \\
&= \sqrt{2} \quad \text{as } x \to \infty
\end{aligned}$$

2.1.5 Indeterminate forms

Some limits such as the form $\frac{\infty}{\infty}$ cannot be detemined in advance and is hence called an indeterminate form. Other indeterminate forms include:

- $\infty - \infty$ if $f(x) \to \infty$ and $g(x) \to \infty$ as $x \to \infty$, $\quad f(x) - g(x) \to ?$
- $\frac{0}{0}$ if $f(x) \to 0$ and $g(x) \to 0$ as $x \to \infty$, $\quad \frac{f(x)}{g(x)} \to ?$
- $0 \times \infty$ if $f(x) \to 0$ and $g(x) \to \infty$ as $x \to \infty$, $\quad f(x)g(x) \to ?$

2.2 The definition of $\lim_{x\to\inf} f(x)$

Suppose that $b, L \in \mathbb{R}$, and f is a real-valued function defined on (b, ∞). It is said:

$$\lim_{x\to\infty} f(x) = L$$

if for every $\epsilon > 0$, there is $M \in \mathbb{R}$ such that:

$$x > M \Rightarrow |f(x) - L| < \epsilon$$

2.3 Proving that $\lim_{x\to\infty} f(x) = L$ using the limit definition

Example

Prove that $\lim_{x\to\infty} \frac{5x}{x+3} = 3$.

$$
\begin{aligned}
&|f(x) - L| < \epsilon \text{ if ?} \\
&|f(x) - L| = \left|\frac{5x}{x+3} - 5\right| = \left|\frac{5x - 5(x+3)}{x+2}\right| = \left|\frac{-15}{x+3}\right| \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad = \frac{15}{x+3} \quad \text{for } x > 3 \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad < \frac{15}{x} \quad \text{for } x > 0 \\
&\Rightarrow |f(x) - L| < \epsilon \quad \text{if} \quad \frac{15}{x} < \epsilon \text{ and } x > 0 \\
&\qquad \text{(i.e. } x > \frac{15}{\epsilon} > 0) \\
&\Rightarrow \text{For any } \epsilon > 0, \exists M = \frac{15}{\epsilon} \text{ such that} \\
&\qquad |f(x) - L| < \epsilon \text{ whenever } x > M
\end{aligned}
$$

Strategy for finding M

1. Find a good upper bound for $|f(x) - L|$
2. Find a simple condition on x such that this upper bound is less than ϵ
3. Use this condition to state an appropriate value for M

Note: The value of M is not unique. If M_0 is a value for M, then any upper bound of M_0 is also a value for M

Example

Show that $\lim_{x\to\infty} \frac{x^2 - \cos x}{x^2+1} = 1$

$$
\begin{aligned}
\left|\frac{x^2 - \cos x}{x^2+1} - 1\right| = \left|\frac{x^2 - \cos x - (x^2+1)}{x^2+1}\right| = \left|\frac{-\cos x - 1}{x^2+1}\right| &= \left|\frac{\cos x + 1}{x^2+1}\right| \\
&\le \frac{2}{x^2+1} \\
&\le \frac{2}{x^2}
\end{aligned}
$$

$$
\begin{aligned}
&\Rightarrow |f(x) - 1| < \epsilon \quad \text{if} \quad \frac{2}{x^2} < \epsilon \\
&\qquad \because \frac{2}{x^2} < \epsilon \Leftrightarrow \sqrt{\frac{2}{\epsilon}} < x \quad \text{for} \quad x > 0, \quad M \triangleq \sqrt{\frac{2}{\epsilon}}
\end{aligned}
$$

2.4 Proofs of basic limit results

Not assessable.

2.5 Limits of functions at a point

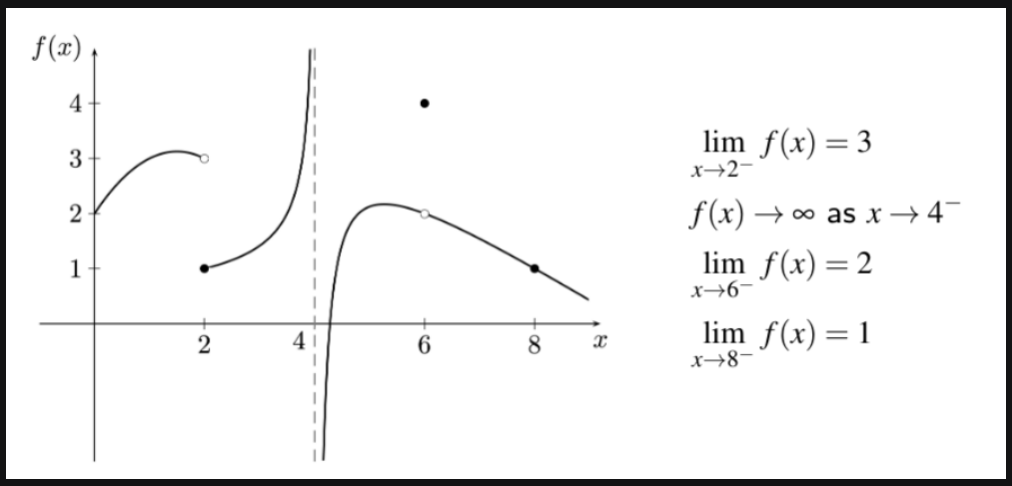
2.5.1 Left-hand, right-hand and two-sided limits

- If $f(x)$ approaches L as x approaches a from the left-hand side, then:

$$\lim_{x \to a^-} f(x) = L \qquad \text{(left-hand limit)}$$

- If $f(x)$ gets arbitrarily large as x approaches a from the left side, then:

$$f(x) \to \infty \quad \text{as} \quad x \to a^-$$

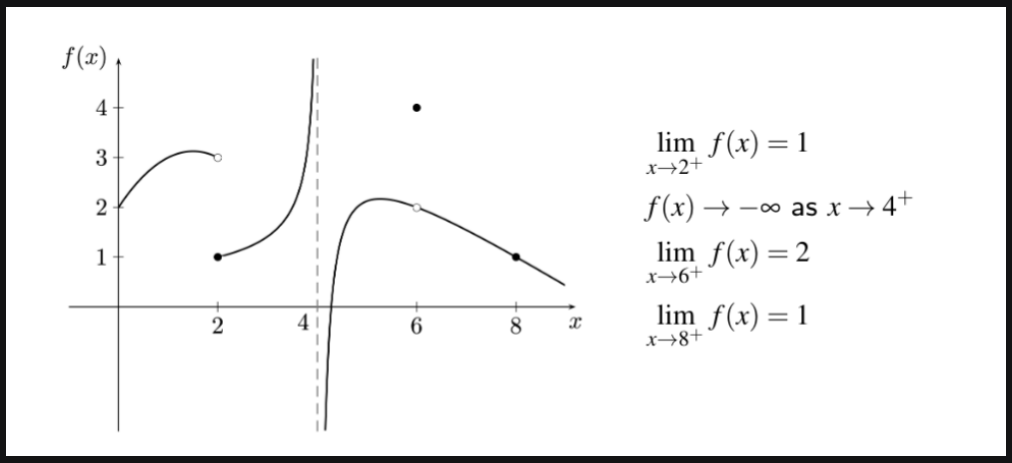


- If $f(x)$ approaches L as x approaches a from the right-hand side, then:

$$\lim_{x \to a^+} f(x) = L \qquad \text{(right-hand limit)}$$

- If $f(x)$ gets arbitrarily large as x approaches a from the right side, then:

$$f(x) \to \infty \quad \text{as} \quad x \to a^+$$



- Left I be an open interval that contains the point a. For a real-valued function f defined on I except possibly at a, if both one-sided limits exist and

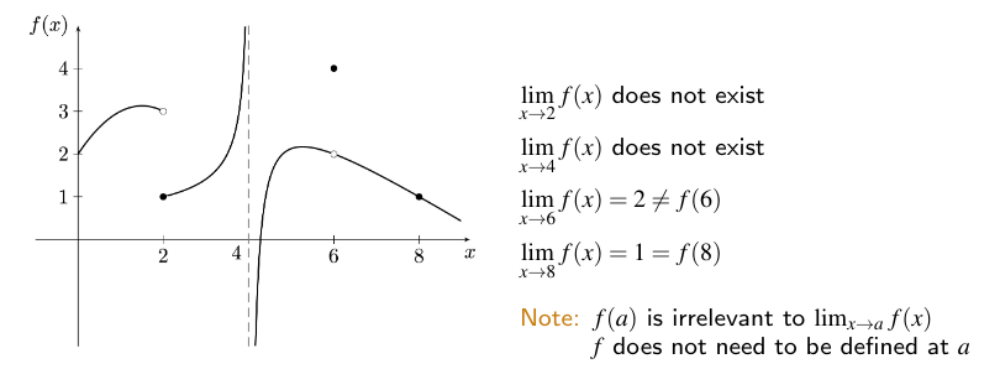
$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L,$$

then we say that thw (two-sided) limit of $f(x)$ as $x \to a$ exists and is equal to L, and we write

$$\lim_{x\to a} f(x) = L.$$

Otherwise, $\lim_{x\to a} f(x)$ does not exist.

- Note: $f(a)$ is irrelevant to $\lim_{x\to a} f(x)$. f does not need to be defined at a



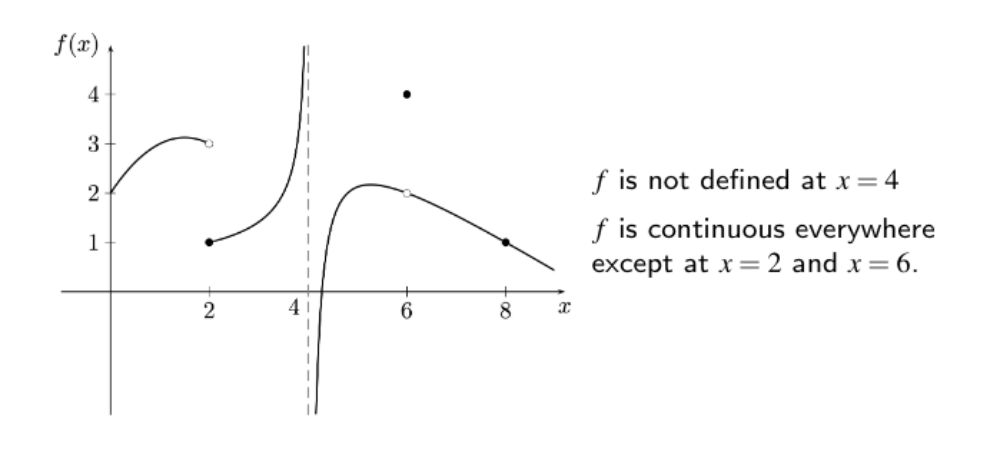
2.5.2 Limits and continuous functions

Let I be an open interval that contains the point a. A function $f : I \to \mathbb{R}$ is continuous at a if

$$\lim_{x\to a} f(x) = f(a)$$

Otherwise, we say that f is discontinuous at a.

If f is continuous at every point of $\mathrm{Dom}(f)$, then we say that f is continuous everywhere.



2.5.3 Rules for limits at a point

Arithmetic

Suppose that $a \in \mathbb{R}$, and $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} g(x) = L_2$. Then

- $\lim_{x\to a}(f(x)+g(x)) = L_1 + L_2$
- $\lim_{x\to a}(f(x)-g(x)) = L_1 - L_2$
- $\lim_{x\to a} f(x)g(x) = L_1 L_2$
- $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L_1}{L_2}$

Function composition

If $\lim_{x\to a} f(x) = L$ and g is continuous at L, then

$$\lim_{x\to a} g(f(x)) = g\left(\lim_{x\to a} f(x)\right) = g(L)$$

Example

Find the limit $\lim_{x\to 5}\sqrt{2x+\cos\pi x^2}$.

$$\begin{aligned}
\lim_{x\to 5}\sqrt{2x+\cos\pi x^2} &= \sqrt{\lim_{x\to 5} 2x+\cos\pi x^2} \qquad \left(\sqrt{} \text{ is continuous}\right) \\
&= \sqrt{\lim_{x\to 5} 2x + \lim_{x\to 5}\cos\pi x^2} \\
&= \sqrt{\lim_{x\to 5} 2x + \cos\lim_{x\to 5}\pi x^2} \qquad (\cos \text{ is continuous}) \\
&= \sqrt{10+\cos 5^2\pi} \qquad \left(\cos 5^2\pi = -1\right) \\
&= 3
\end{aligned}$$

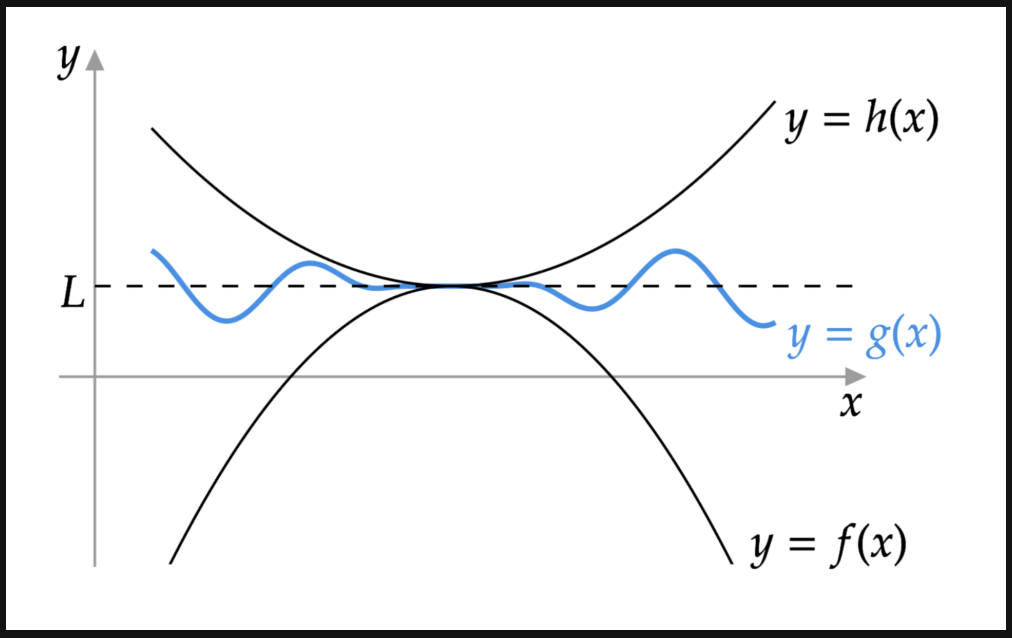
Pinching theorem (two-sided limit)

Let I be an open interval that contains the point a. Suppose that f, g, and h are functions defined on I. If

- $f(x) \le g(x) \le h(x), \quad \forall x \in I$, and
- $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$

then

$$\lim_{x\to a} g(x) = L$$



Example

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is defined by the formula

$$f(x) = \begin{cases} \frac{|x^2-9|}{x-3} & \text{if } x \neq 3 \\ -6 & \text{if } x = 3 \end{cases}$$

Discuss the limiting behaviour of $f(x)$ as x approaches 3.

Limit from left hand side

$$\text{For } x \in (-3,3), \qquad \frac{|x^2-9|}{x-3} = \frac{|x+3||x-3|}{x-3} = -(x+3)$$
$$\Rightarrow \lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} -(x+3) = -6 = f(3)$$

Limit from right hand side

$$\text{For } x > 3, \qquad \frac{|x^2-9|}{x-3} = x+3$$
$$\Rightarrow \lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} x+3 = 6$$

Since $\lim_{x\to 3^+} f(x) \neq \lim_{x\to 3^-} f(x)$, two-sided limit does not exist at $x = 3$
So, f is not continuous at $x = 3$

Chapter 3

Properties of continuous functions

3.1 Combining continuous functions

Suppose that the functions f and g are continuous at a point a. Then

1. $f+g$, $f-g$ and fg are continuous at a
2. $\frac{f}{g}$ is continuous at a, provided that $g(a) \neq 0$

$$\begin{aligned}
\lim_{x\to a}(f+g)(x) &= \lim_{x\to a}(f(x)+g(x)) && \text{(by definition of } f+g\text{)} \\
&= \lim_{x\to a} f(x) + \lim_{x\to a} g(x) && \text{(by "limit of sum = sum of limits")} \\
&= f(a)+g(a) && \text{(by continuity of } f \text{ and } g \text{ at } a\text{)} \\
&= (f+g)(a)
\end{aligned}$$

Supose that the function f is continuous at a, and g is continuous at $f(a)$. Then $g \circ g$ is continuous at a.

$$\begin{aligned}
\lim_{x\to a}(g\circ f)(x) &= \lim_{x\to a} g(f(x)) \\
&= g\left(\lim_{x\to a} f(x)\right) && \text{(by limit rule of function composition)} \\
&= g(f(a)) && \text{(by continuity of } f \text{ at } a\text{)} \\
&= (g\circ f)(a)
\end{aligned}$$

Example

Every polynomial $p : \mathbb{R} \to \mathbb{R}$ is continuous at every $a \in \mathbb{R}$.

$$\begin{aligned}
\Leftarrow \text{for any } n \in \mathbb{N}, \quad p(x) &= a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \\
&= \text{product \& sum of } g(x) = x \text{ and contant functions, ie. continuous by (i)}
\end{aligned}$$

3.2 Continuity on intervals

For a real-valued function f defined on an open interval (a,b), we say that f is continuous on (a,b) if it is continuous at every $x \in (a,b)$.

For a real-valued function f defined on a closed interval $[a,b]$, we say that

- f is continuous at the endpoint a if $\lim_{x \to a^+} f(x) = f(a)$
- f is continuous at the endpoint b if $\lim_{x \to b^-} f(x) = f(b)$
- f is continuous on $[a,b]$ if it is continuous on (a,b) and at endpoints a and b.

3.3 The intermediate value theorem

Suppose that f is continuous on $[a,b]$, and $f(b) < z < f(a)$ for some z. The graph of f from $(a, f(a))$ to $(b, f(b))$ without lifting the pen. We must cross a point (c,z) along the way for some $c \in (a,b)$.

Let $f : [a,b] \to \mathbb{R}$ be a continuous function. If either $f(a) < z < f(b)$ or $f(b) < z < f(a)$ for some $z \in \mathbb{R}$, then there is at least one $c \in (a,b)$ such that $f(c) = z$

- In general, c is not unique
- The continuity of f is necessary
- Only for continuous functions defined on (subsets of) $\mathbb{R}$, no analogue for $\mathbb{Q}$

$$\begin{aligned}
&f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2 - 2 && f \text{ is continuous on } \mathrm{Dom}(f) = \mathbb{R} \\
& && = \in (f(0), f(2)),\ f(c) = 0 \text{ at } c = \sqrt{2} \in (0,2) \\
&f : \mathbb{Q} \to \mathbb{Q}, \quad g(x) = x^2 - 2 && g \text{ is continuous on } \mathrm{Dom}(g) = \mathbb{Q} \\
& && = \in (g(0), g(2)),\ g(c) = 0 \text{ at } c = \sqrt{2} \notin \mathbb{Q}
\end{aligned}$$

- Since infinity is not part of $\mathbb{R}$, a definite bound must be defined before using IVT

Example

Show that there exists a solution $c \in [1,2]$ of the equation $\sqrt{c} = c^2 - 1$ and approximate its value.

Define a function $f : [1,2] \to \mathbb{R}$ by $f(x) = x^2 - 1 - \sqrt{x}$. Then,
- f is continuous on [1,2]
- $f(1) = -1$ and $f(2) = 3 - \sqrt{2} > 0$.

Since $f(1) < 0 < f(2)$, by IVT, there is a number $c \in (1,2)$ such that $f(c) = c^2 - 1 - \sqrt{c} = 0$

Approximation:
- take $c_1 = 1.5 \Rightarrow f(c_1) = 0.0253$, by IVT, $\exists$ a solution $c \in (1, c_1)$
- take $c_2 = 1.25 \Rightarrow f(c_2) = -0.5555$, by IVT, $\exists$ a solution $c \in (1, c_2)$

3.4 The maximum-minimum theorem

Let f be a real-valued function defined on a closed interval $[a,b]$.

- A point $c \in [a,b]$ is a global minimum point for f on $[a,b]$ if

$$f(c) \le f(x), \quad \forall x \in [a,b].$$

- A point $d \in [a,b]$ is a global minimum point for f on $[a,b]$ if

$$f(d) \ge f(x), \quad \forall x \in [a,b].$$

If f is continuous on $[a,b]$, then f attains its minimum and maximum on $[a,b]$, ie. there exists points $c,d \in [a,b]$ such that

$$f(c) \leq f(x) \leq f(d), \quad \forall x \in [a,b].$$

- Closed intervals are necessary
- Continuity of f is necessary

Example

Let f be a function defined by $f(x) = \begin{cases} \frac{1}{x^2+1} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

1. Is f bounded on $[-2,2]$?

$$\text{Yes.} \quad \text{For } x \neq 0, \quad |f(x)| = \left| \frac{1}{x^2+1} \sin \frac{1}{x} \right| \leq \frac{1}{x_1^2} \leq 1.$$

$$\text{For } x = 0, \quad f(x) = 1.$$

$$\Rightarrow |f(x)| \leq 1, \quad \forall\, x \in [-2,2]$$

2. Does it attain a minimum or maximum on $[-2,2]$
   - f on $[-2,2]$ attains a maximum at $x = 0$.
   - f does not have a minimum value on $[-2,2]$

$$\text{For } x_n = \left(\frac{-\pi}{2} + 2n \right)^{-1}, \quad n \in \mathbb{Z}$$

$$\sin \frac{1}{x_n} = -1$$

$$f(x_n) = -\frac{1}{\left(\frac{-\pi}{2} + 2n\pi\right)^{-2} + 1}$$

Then as $n \to \infty$, $\quad x_n \to 0$ and $f(x_n) \to -1$

Note: $f(x) > -1$

$\Rightarrow$ as $x \to 0, f(x)$ oscillates between (not touching) -1 and 1